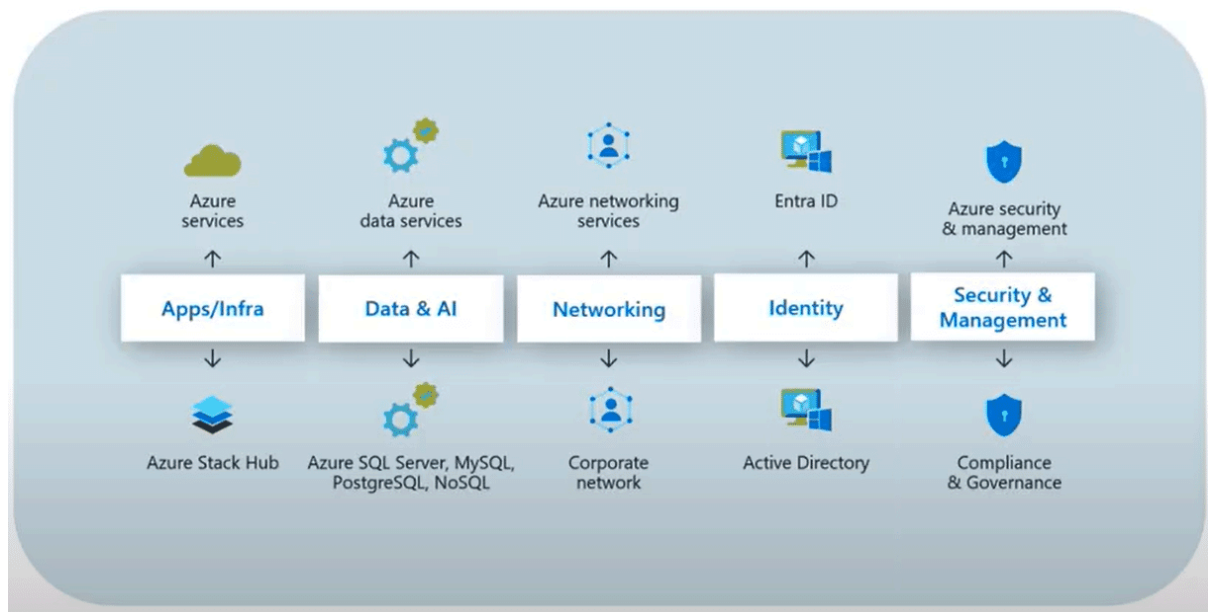


Azure is a cloud computing platform and online portal that allows you to access and manage cloud services and resources provided by Microsoft. These services and resources include storage and data transformation, depending on your needs. To access these resources and services, all you need is an internet connection and the ability to connect to the Azure portal.



Things you need to know about Azure:

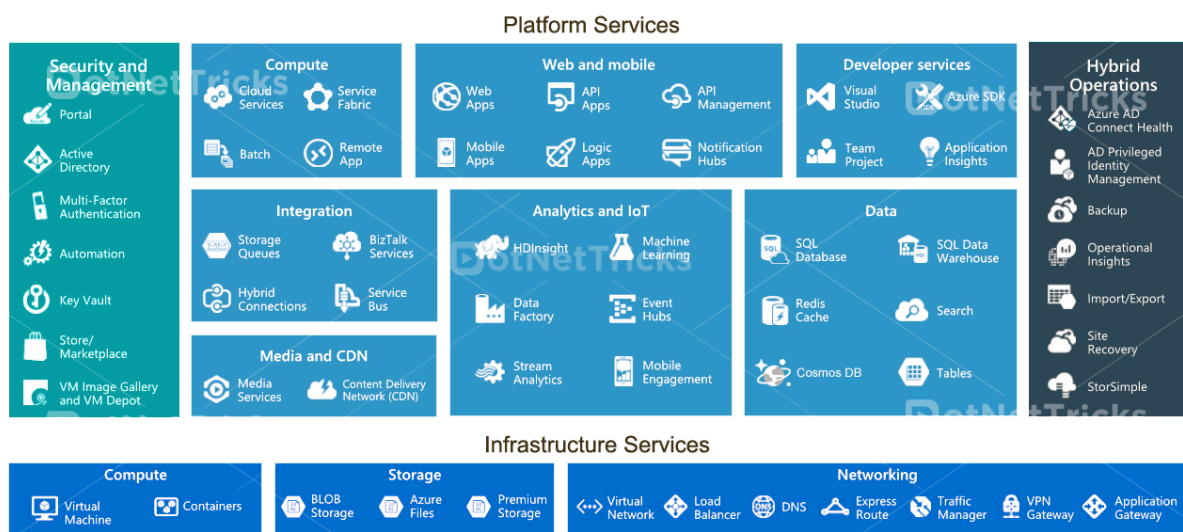
- Azure was released on February 1, 2010, significantly later than its main competitor, AWS.
- Azure follows a free-to-start model, meaning you only pay for the services you choose to run on Azure.
- 80% of the Fortune 500 companies use Azure for their cloud computing needs.
- Azure supports multiple programming languages, including Java, Node.js, and C#.
- Another advantage of Azure is the number of data centers available worldwide. There are 42 (and still growing) Azure data centers spread globally, the highest

number of data centers for any cloud platform. In addition, Azure is planning to open 12 more data centers, bringing the total to 54.

Why should you use Azure?

- Application development You can create any web application in Azure.
- Testing After developing your application, you can test it directly on Azure.
- Application hosting After testing is complete, Azure can help you host your applications.
- Create virtual machines You can create virtual machines in any configuration you want with the help of Azure.
- Integrate and sync features Azure allows you to integrate and synchronize virtual devices and folders.
- Collect and store metrics Azure allows you to collect and store metrics, which can help you find what's performing well.
- Virtual hard drives These are extensions of virtual machines; they provide a huge amount of data storage.

Azure services



Azure offers over 200 services, divided into 18 categories, including: **computing, networking, storage, IoT, migration, mobile, analytics, containers, artificial intelligence, machine learning, integration, management tools, developer tools, security, databases, DevOps, media identity and web services**. Let's take a look at some of the main Azure services categorized by VinSEP:

Computing

- **Virtual Machine** This service allows you to create a virtual machine in Windows, Linux, or any other configuration in seconds.
- **Cloud Service** This service allows you to create scalable applications in the cloud. Once the application is deployed, everything, including load balancing and health monitoring, is handled by Azure.
- **Service Fabric** With this service, the process of developing a microservice is greatly simplified. A microservice is an application that contains other smaller applications packaged together.
- **Functions** With Functions, you can create applications using any programming language. The best part about this service is that you don't need to worry about hardware requirements while developing your application because Azure will take care of that. All you need to do is write the code.

Networking

- **Azure CDN** Azure CDN (Content Delivery Network) helps deliver content to users, utilizing high bandwidth and enabling content to be delivered to anyone globally. CDN services utilize a network of strategically placed servers worldwide so users can access data as quickly as possible.
- **Express Route** This service allows you to connect your on-premises network to the Microsoft cloud or any other service you want, via a private connection.

Therefore, the only communication that will occur here will be between the enterprise network and the service you wish to connect to.

- **Virtual network**A virtual network allows any Azure service to communicate with each other privately and securely.
- **Azure DNS**This service allows you to host DNS domains or system domains on Azure.

Storage

- **Disk Storage**This service allows you to choose between HDD or SSD as your storage option along with your virtual machine.
- **Blob Storage**This service is optimized for storing large amounts of unstructured data, including text and even binary data.
- **File Storage**This is a managed file storage service that can be accessed via the SMB (server message block) protocol.
- **Queue Storage**With this service, you can provide stable message queuing for a large workload. This service can be accessed from anywhere in the world.